



Rijksinstituut voor Volksgezondheid
en Milieu
*Ministerie van Volksgezondheid,
Welzijn en Sport*

National Heat Plan version 2015

RIVM Letter report 2014-0051
WI Hagens | M. van Bruggen



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Colophon

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Werner Hagens (VLH),
Mark van Bruggen (VLH),

Contact:

Werner Hagens
werner.hagens@rivm.nl

This research was carried out on behalf of VWS, in the context of V/200114 'Loket cGM'.

This is a publication of:

National Institute for Health and Environment

PO Box 13720 BA Bilthoven
www.rivm.nl

Audience summary

The National Heat Plan, which was drawn up in 2007 at the request of the Ministry of Health, Welfare and Sport, has been used every year since then. The evaluations have identified several points for improvement that were not yet included in the text of the Heat Plan. The RIVM has now updated the text.

The Heat Plan now better indicates the responsibilities of the various organizations during a period of persistent heat. In collaboration with the KNMI, the accuracy with which persistent heat is predicted has also been improved, because the criteria for the alarm are now set to four days of heat instead of five. In addition, the practical information about measures has been transferred to the new GGD guideline Health risks of summer conditions and the associated Heat toolkit.

Persistent heat poses a risk to the health of certain groups of people, such as the elderly, people in healthcare institutions, chronically ill and overweight people. These risks range from mild symptoms such as fatigue and itching to serious conditions with potentially life-threatening consequences.

The aim of the National Heat Plan is to inform organizations in a timely manner that a period of persistently warm weather is expected. These include GGDs, trade organizations and the Dutch Red Cross. They then warn their supporters and regional contacts, who are provided with measures via the Heat Plan that can limit the consequences of the persistent heat.

Keywords: heat, heat plan, VWS, health risks, measures

Abstract

The Dutch National Heatwave Plan (Nationaal Hitteplan) that was developed in 2007, at the request of the Ministry of Health, Welfare and Sport (VWS), has been used every year since. Evaluations pointed out where the National Heatwave Plan could be improved, and RIVM has updated the text accordingly.

The updated National Heatwave Plan more clearly states the various responsibilities of the different organizations during prolonged periods of excessively hot weather. The level of accuracy in predicting such weather conditions has also been improved, in collaboration with the Royal Netherlands Meteorological Institute (KNMI), with warning criteria having been set at four days of continuous heat, instead of five. In addition, the practical information on measures has been transferred to the new guideline of the Dutch Public Health Service (GGD) on summer-related health risks and the accompanying toolkit 'Hitte' ('heat').

Continuing hot weather poses a health risk for certain groups of people, such as the elderly, people in care institutions, and chronically ill and overweight people.

Risks vary from light symptoms, such as fatigue or itchiness, to severe complaints with possibly life-threatening consequences.

The purpose of the National Heatwave Plan is to provide organizations with a timely warning in case of an expected prolonged period of hot weather.

Organizations include GGD branches, health organizations and the Dutch Red Cross. They will then warn their target groups and regional contacts, who will be provided through the Heat Plan with measures to limit the impact of such hot weather conditions.

Keywords: heatwave, heatwave plan, VWS, health risks, measures

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1 Reason for updating the National Heat Plan

At the end of 2006, the Ministry of Health, Welfare and Sport took the initiative to draw up a National Heat Plan in the Netherlands, which was completed in early 2007. Since 2007, the National Heat Plan has been activated almost every year. After each activation of the National Heat Plan, various organizations (including GGD, KNMI, the Dutch Red Cross) examined how the National Heat Plan could be improved. The criteria for persistent heat have been optimized and it is now also possible to activate the National Heat Plan regionally.

Since 2007, the RIVM has made several products for GGDs and other healthcare professionals. Because this substantive information has become available, it is not necessary to include this information in the National Heat Plan. For substantive information, please refer to the two products below: RIVM toolkit Heat: This toolkit for public communication about

topic Heat contains ready-made products, such as an information brochure, question and answer lists and standard texts (website texts and letters) that can be used by professionals in their communication with their target group (RIVM toolkit Heat, 2014).

RIVM Report: Health risks of summer conditions. This
The GGD guideline was written by a working group consisting of employees from the GGD, RIVM, KNMI and TNO (RIVM 2012).

The updated National Heat Plan version 2015 has been adapted to the optimized working method. The agreements made in 2007 (between the various institutes and institutions) have not been adjusted. Chapter 2 contains the intended goal of the National Heat Plan and an overview of the parties involved. Chapter 3 discusses the method of the plan. For substantive information and practical advice, please refer to the two products above.

2 The National Heat Plan

2.1 Problem sketch

Hot weather poses a health threat to certain people.

The elderly are the largest risk group, because their temperature regulation is generally poorer and the feeling of thirst is reduced. Not only the elderly, but also people with a chronic condition, people in social isolation, the homeless, overweight people and (very) young children are at health risk. The use of certain medicines, alcohol or drugs can also increase the risk of health effects in persistent heat.

Hot weather can lead to complaints such as fatigue, concentration problems, dizziness and headaches. In addition, skin problems such as itching and rash with blisters may occur. In serious situations, dehydration can cause cramps, nausea, exhaustion, faintness and unconsciousness. Dehydration, especially in the elderly and people with cardiovascular disease, can lead to disruption of the blood flow of vital organs (heart, lungs, brain, kidneys) with potentially life-threatening consequences.

Climate studies indicate that a gradual increase in temperatures is likely in the coming decades. Extreme weather events such as heat waves are also likely to occur more often. It therefore appears that activation of the Heat Plan will continue to be necessary in the future.

2.2 Purpose of the Plan

The National Heat Plan is aimed at providing early warning that persistent hot weather is forecast, so that measures can be taken to reduce and (where possible) prevent nuisance caused by persistent heat. Experiences in other countries show that extra attention from emergency services and greater involvement of the environment during periods of persistent heat contributes to the well-being and health of risk groups. The National Heat Plan is in fact a communication plan in which information must reach risk groups and their immediate environment in a timely and complete manner. This environment includes the institutions with which the vulnerable group has contact, but other healthcare providers and volunteers are also part of the environment. This awareness and knowledge are a prerequisite for adequate action during a period of persistent heat.

2.3 Involved parties

The National Heat Plan was drawn up in 2007 in collaboration between the Ministry of Health, Welfare and Sport, the RIVM, the KNMI, the Dutch Red Cross, GGD Netherlands and GHOR-Netherlands. Since 2007, discussions have been held with various (trade) organizations, such as the General Dutch Association for the Elderly (ANBO), ActiZ (trade organization of healthcare entrepreneurs), TNO, the National Association of General Practitioners (LHV), the Nurses & Caregivers Netherlands (V&VN) and the Royal Dutch Society for the Promotion of Pharmacy (KNMP) to actively involve them in the National Heat Plan.

Below is a description of the organizations involved and their role in the National Heat Plan.

2.3.1 *The Ministry of Health, Welfare and Sport*

In 2006, the Ministry of Health, Welfare and Sport took the initiative to draw up a National Heat Plan in the Netherlands. The RIVM was also commissioned to update the National Heat Plan in 2014 by the Ministry of Health, Welfare and Sport.

2.3.2 *KNMI*

The KNMI makes the weather forecast and calculates (based on these weather forecasts) the chance that a period of persistent heat is expected. This is based on the definition of "persistently warm weather" described in the next chapter. If there is a high chance of persistent warm weather, the KNMI will report this to the RIVM. The KNMI can issue a weather warning for extreme heat ("code yellow") for one or more regions, which can be seen on the website: www.knmi.nl.

2.3.3 *RIVM*

The RIVM evaluates the messages from the KNMI about the risk of persistent heat and, together with the KNMI, decides to send a preliminary warning or a warning to all registered recipients of the heat message (including GGDs, trade organizations and the Dutch Red Cross). The RIVM manages the mailing list. National organizations that are interested in also receiving the heat reports can register for this list by sending an email to cGM@rivm.nl. Regional and local organizations can contact the local GGD for this.

The RIVM has made a number of products for GGDs and other healthcare professionals (RIVM, 2012; RIVM toolkit Hitte, 2014). These contain information about the health effects of warm weather conditions. These products also contain useful advice on how to limit nuisance caused by heat. In addition, RIVM answers questions from GGDs and professionals (not from the general public) about heat and health. Questions can be asked to RIVM by sending them to cGM@rivm.nl. Questions from the general public can be answered by the GGDs.

2.3.4 *GGD GHOR Netherlands*

The most important task of GGD GHOR Netherlands consists of collectively representing the interests of the local and regional GGD and GHOR towards politics, (local) governments, insurers, partners, education, media and the public.

2.3.5 *Local GGDs*

Municipalities are responsible for the well-being of their citizens, through the Public Health Act. This law states that identifying undesirable situations, advising the population about risks, answering questions from the population and providing information is a legal task (Public Health Act, 2008). The GGD can support municipalities in carrying out this task.

The role of local GGDs in the National Heat Plan is primarily to advise municipalities on health risks and how to communicate about them.

In addition, GGDs can inform the general public, (care) institutions and healthcare providers about the risks of persistent heat. Practical advice can be given on how to limit the nuisance caused by the heat

become. The practical implementation for GGDs is described in the GGD guideline Medical Environmental Science: Health risks of summer conditions (RIVM, 2012). It is up to the local GGD itself to shape the local implementation of the National Heat Plan.

2.3.6 *The Dutch Red Cross*

The Dutch Red Cross works with 33,000 volunteers. They work in their own hometown or region for 254 local departments in the Netherlands. The Red Cross provides assistance throughout the country through these departments. In the event of a warning from the National Heat Plan, the Dutch Red Cross can share knowledge and practical instructions with volunteers and other affiliated organizations. The way the Dutch Red Cross is organized gives it insight into local aid needs, allowing it to act quickly during heat.

2.3.7 *Branch organisations*

In recent years, discussions have taken place with various industry and interest groups in the healthcare chain. These include the General Dutch Association for the Elderly (ANBO), the National Association of General Practitioners (LHV), ActiZ (organization of healthcare entrepreneurs), the Royal Dutch Society for the Promotion of Pharmacy (KNMP) and the Nurses & Caregivers Netherlands (V&VN). The RIVM has shared information with these interest groups, so that these organizations can create products themselves, for example for newsletters to their members or information on their own website. These organizations receive heat warnings from the RIVM and can pass this information on to their supporters and draw their attention to the risks of persistent heat. Through their own channels they promote a sense of urgency, knowledge and professional action in times of persistent heat. However, it is up to the organizations to organize this themselves.

2.3.8 *Healthcare institutions*

Healthcare institutions also prepare for periods of persistent heat by identifying in advance the measures they will take to guarantee the quality of care during this period. If such a period occurs, these measures will be implemented. However, it is up to the organizations themselves to organize this. For example, ActiZ, as an organization of healthcare entrepreneurs, has guidelines on its website to optimize care during heat.

There is no obligation for institutions to draw up a specific heat protocol, partly because the situation and circumstances can differ greatly from institution to institution. However, all healthcare providers are obliged to provide good care. Taking adequate measures in extreme heat is an explicit part of this. The Healthcare Inspectorate (IGZ) monitors this (Answering parliamentary questions, 2013).

2.3.9 *Media*

The media can disseminate information at its own discretion about the occurrence of a period of persistent heat, the problems this may pose for certain groups and the measures that can be taken to limit the negative effects of the heat on health. In 2013 it turned out that the National Heat Plan had been taken up carefully and with sufficient attention by the (national) media.

3 Implementation of the National Heat Plan

3.1 Definition of a period of persistent heat

The criterion for persistent heat is an expectation of a period of four or more days with a maximum temperature above 27 °C. This definition of a period of persistent heat is different from the definition for a heat wave (at least 5 consecutive summer days (25 °C or higher), of which at least 3 tropical days (30 °C or higher), measured in De Bilt. A heat wave can only be determined afterwards. be determined, while the National Heat Plan must predict in advance that it will be persistently warm.

The reasons why persistent heat was chosen for this criterion are: 1. do not occur too often in the Netherlands. When it does occur, it is usually really warm. 2.

The probability of occurrence of a period of 4 days with a Temperatures above 27 °C can be estimated well in advance (unlike a heat wave).

NB: After each activation of the National Heat Plan, an evaluation follows to optimize the working method. It may therefore be that an adapted definition, or the use of additional values, such as perceived temperature or the use of the temperature at night, can be used in the future if it turns out that this improves the predictability of the warm period and effectiveness of the National Heat plan, will be enlarged.

3.2 Division into phases

The National Heat Plan is divided into three phases.

- Vigilance phase;
- Pre-warning phase;
- Warning phase.

3.2.1 *Vigilance phase*

The first phase of vigilance is determined by the onset of the summer period. This phase is in effect from June 1 to September 1. The significance of this phase is that organizations and institutions realize before the summer that they have to prepare for a period of persistent heat. In this phase, for example, internal agreements regarding heat can be checked and updated if necessary.

3.2.2 *Pre-warning phase*

The second phase that can occur is that of the advance warning. If the chance of a period of four or more days with a maximum temperature above 27 °C is small, but present, the RIVM will be informed of this by the KNMI. A pre-warning message will then be forwarded to all contact persons of the heat mailing. The significance of this advance warning is that the receiving organizations are alert in case a period of persistent heat occurs. The organizations can check internally whether they are sufficiently able to fulfill their agreed tasks if the predicted heat continues. The range of tasks that must be completed differs per type of organization. It is your own responsibility

every organization to initiate the correct actions based on the advance warning. The general public is not intended to be warned at this stage. The chance of a period of persistent heat is still too small for that. Warning the general public too early and too frequently does not contribute to the credibility of the system.

3.2.3

Warning phase

As soon as the KNMI reports that there is a high chance of a period of four or more days with a maximum temperature above 27 °C, consultations will take place between the RIVM, the KNMI safety meteorologist and the communications departments of both organizations about the need to issue the advance warning. to transition into a warning phase. If it is decided to enter the warning phase, the National Heat Plan is activated. This is usually known more than 24 hours in advance.

The warning system of the National Heat Plan consists of a stepped information chain (see figure 1). In coordination with the KNMI, the RIVM sends a warning email with informative content and current weather information to all recipients of the heat message (including GGDs, trade organizations and the Dutch Red Cross). It is up to these recipients to use the messages and forward them to their own supporters and regional contacts.

An update is sent daily. The Dutch population is also informed by a press release issued by the RIVM and the KNMI. The weather reports on the KNMI website will contain a weather warning for extreme heat ("code yellow") for one or more regions.

In this phase, the various organizations can take the measures they have determined in advance. For example, a healthcare institution can implement its own heat plan and a trade organization can post a message on the website or forward a news item to their supporters. For example, information about the risks of warm weather is also posted on the RIVM website, with clear and practical advice to limit nuisance.

The warning phase ends with a clear email to all recipients of the heat message stating from which day the National Heat Plan is no longer active.

3.3

Warnings by province

In the Netherlands it is not equally warm everywhere. In the past, it has happened that the National Heat Plan was activated because it was very warm in the East and South of the country, while the temperatures in the North and the coastal provinces were not extreme. Significant regional differences in temperature may occur. It has therefore been possible for several years to activate the National Heat Plan per region (which largely corresponds to the provinces).

3.4

Events

At events, large groups of people can be at increased risk due to a combination of factors. The risks depend on the type of event (e.g. physical activity) and the target group (e.g. elderly). In addition, both personal behavior and environmental characteristics play an important role.

Many municipalities have a safety plan that contains general conditions that an event must meet. This may also include measures for extreme weather conditions. Event organizers can contact the GGD and/or the GHOR in the region in advance to identify the risks of extreme weather conditions. The GGD can also advise the organizers directly or through the municipality for smaller events such as a school trip, sports day or neighborhood activity. The GGD guideline for medical environmental science: Health risks of summer conditions (RIVM, 2012) describes a practical approach for support at events.

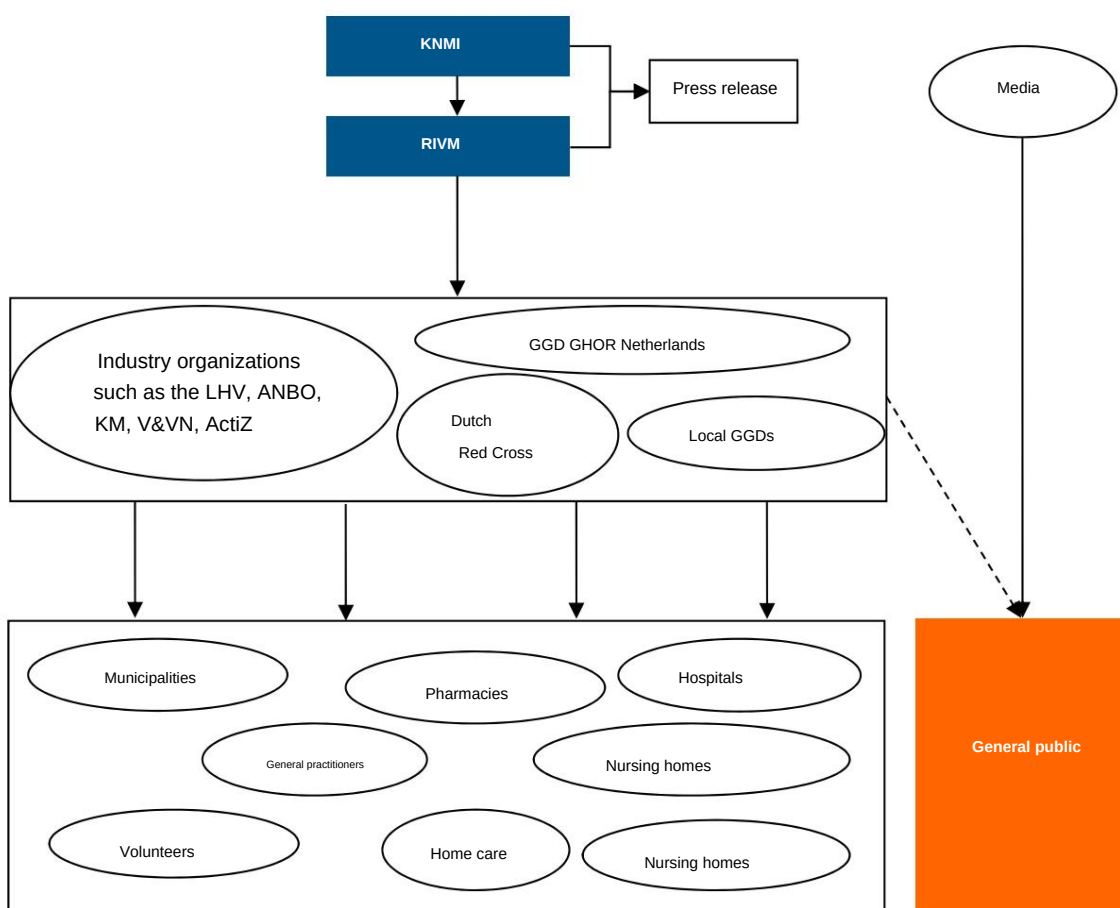


Figure 1: Schematic representation of the stepped information chain in the warning phase. The KNMI provides the information to the RIVM, after which the RIVM, in coordination with the KNMI, sends a warning email to all recipients of the heat message (including GGDs, trade organizations and the Dutch Red Cross). It is up to these recipients to use the messages and forward them to their own supporters and regional contacts.

The KNMI and the RIVM prepare a (joint) press release and provide information via the website. The media can disseminate information at its own discretion. Local GGDs, trade organizations and the Dutch Red Cross can also provide the general public with information.

3.5 Crisis situation

If, in addition to the persistent heat, other risk situations occur, such as water scarcity or power outages, it can be scaled up to a crisis situation. The existing rules and agreements from the (regional) crisis structure apply. This is not taken into account in the National Heat Plan.

4 Information sources

Answer to parliamentary questions (2013) Answer to parliamentary questions about the report that elderly care is ill-prepared for a heat wave. Accessed October 2014. <http://www.rijksoverheid.nl/documents-en-publicaties/kamerstukken/2013/07/26/beantworing-kamerquestions-over-het-bericht-dat-de-ouderenzorg-bad-voor-prepared-is-on-a-heatwave.html>

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Public Health Act (2008) Act of 9 October 2008, containing provisions on public health care. http://wetten.overheid.nl/BWBR0024705/geldigheidsdatum_29-10-2014

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